STAT6611

- Lecture 10 (R, Sept 24)
- solutions posted to Homework 1
- Homework 2 assigned due F, Oct 2 by midnight (next week)

practice on LSE (without constraints)

- My office hours: TW 12:30-1:30 PM
  & R 1-2 PM

TA Ben: R 9:30-10:30 AM
in Snedecor 2207

Summary of Least Squares Estimation ($\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, $E(\underline{\varepsilon}) = \underline{0}$)

NES

$X'X\hat{\beta} = X'\underline{Y}$ have solution $\Leftrightarrow$ any/only solutions $\hat{\beta}$ minimize

$Q(\beta) \equiv (\underline{Y} - X\beta)'(\underline{Y} - X\beta)$ sum of squares

$\underline{\lambda}'\underline{\beta}$ estimable $\Leftrightarrow \underline{\lambda}' \in R(X) \Leftrightarrow \underline{\lambda}'\hat{\underline{\beta}}$ doesn't change with solution to NEs

$\Leftrightarrow$ conjugate NEs have solution

$(X'X)\underline{r} = \underline{\lambda}$

For estimable $\underline{\lambda}'\underline{\beta}$, LSE $\underline{\lambda}'\hat{\underline{\beta}} = \underline{r}'X'\underline{Y}$ & is BLUE under Model II (Gauss-Markov)

Constraints: $A\underline{\beta} = \underline{b}$ ($A\underline{f} = \underline{b}$ must have solution or $AA^{-}\underline{b} = \underline{b}$)

constrained NEs

$$\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \hat{\underline{\beta}} \\ \underline{f} \end{pmatrix} = \begin{pmatrix} X'\underline{Y} \\ \underline{b} \end{pmatrix}$$ have solution & any/only solutions $\hat{\underline{\beta}}$ (for some $\underline{f}$) minimize

$Q(\beta) \equiv (\underline{Y} - X\underline{\beta})'(\underline{Y} - X\underline{\beta})$ subject to $A\underline{\beta} = \underline{b}$

$\underline{\lambda}'\underline{\beta}$ estimable $\Leftrightarrow \underline{\lambda}' \in R\begin{pmatrix} X \\ A \end{pmatrix} = R\begin{pmatrix} X'X \\ A \end{pmatrix} \Leftrightarrow \underline{\lambda}'\hat{\underline{\beta}}$ doesn't change with solution $\hat{\underline{\beta}}$ to constrained NEs

$\Leftrightarrow$ constrained conjugate NEs have solution

$$\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix} = \begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix}$$

For estimable $\underline{\lambda}'\underline{\beta}$,

constrained LSE $\underline{\lambda}'\hat{\underline{\beta}} = \underline{r}_2'\underline{b} + \underline{r}_1'X'\underline{Y}$ & is BLUE under Model II constraint

Corollary. $X\underline{\widehat{\beta}} = X\underline{\widetilde{\beta}}$ for the first components $\underline{\widehat{\beta}}$ and $\underline{\widetilde{\beta}}$ of any two solutions to the constrained normal equations.

all rows of X are estimable (in $R\begin{pmatrix} X \\ A \end{pmatrix}$)
so $X\underline{\hat{\beta}} = X\underline{\tilde{\beta}}$ holds

The following lemma, which will be useful in the next development, is an immediate consequence of Theorem 2.11 and Lemma 2.12 (i).

Lemma 2.14. *Under Models IIa—VIIa, a linear function* $\underline{\lambda}'\underline{\beta}$ *is estimable if and only if* $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$. $\Leftrightarrow \underline{\lambda}' \in R\begin{pmatrix} X \\ A \end{pmatrix} = R\begin{pmatrix} X'X \\ A \end{pmatrix}$

The equations that play the role of the conjugate normal equations in our development are the constrained conjugate normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix},$$

constrained NEs
$\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \underline{\beta} \\ \underline{t} \end{pmatrix} = \begin{pmatrix} X'\underline{y} \\ \underline{b} \end{pmatrix}$

that is, $X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda}$ and $A\underline{r}_1 = \underline{0}$.

Theorem 2.16. *Under Models IIa—VIIa,* $\underline{\lambda}'\underline{\beta}$ *is estimable, that is,*

$$\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}, \quad \textit{that is,} \quad \underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix},$$

if and only if the constrained conjugate normal equations are consistent.

i.e. some solution $\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix}$ exists

Proof. For the "only if" part ($\Longrightarrow$), suppose that $\underline{\lambda}'\underline{\beta}$ is estimable, or $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$, implying $\underline{\lambda}' = \underline{r}_1'X'X + \underline{r}_2'A$ for some $\underline{r}_1$ and $\underline{r}_2$. It suffices to use Lemma 2.10: take $\underline{a} = \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix}$ to be any vector such that

$$\underline{a}' \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1', \underline{a}_2') \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1'X'X + \underline{a}_2'A, \underline{a}_1'A') = \underline{0}' \quad (*)$$

and show

$$\underline{a}' \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = 0.$$

this shows $\begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix}$ is in $C\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}$ or constrained conjugate NEs have solution

From the proof of Theorem 2.12, (*) implies $\underline{a}_1'X' = 0$ & $\underline{a}_1'A' = 0$.

Then, $\underline{a}'\begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix} = (\underline{a}_1'\ \underline{a}_2')\begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix} = \underline{a}_1'\underline{\lambda}$

$= \underline{a}_1'(X'X\underline{r}_1 + A'\underline{r}_2)$

$\underline{\lambda}'\underline{\beta}$ estimable $\rightarrow$ $= \underbrace{\underline{a}_1'X'}_{0}X\underline{r}_1 + \underbrace{\underline{a}_1'A'}_{0}\underline{r}_2 = 0$

So, $\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix} = \begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix}$ holds for some $\underline{r}_1, \underline{r}_2$

($\Longleftarrow$) if $\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix} = \begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix}$ holds $\Rightarrow \underline{\lambda}' = \underline{r}_1'X'X + \underline{r}_2'A$ holds

$\Rightarrow \underline{\lambda}' \in \mathcal{R}\begin{pmatrix} X'X \\ A \end{pmatrix} \rightarrow \underline{\lambda}'\underline{\beta}$ is estimable

Corollary. Under Models IIa—VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}^{-}\begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$

constr. conjugate NEs have solution $\Leftrightarrow \begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}^{-}\begin{pmatrix} \underline{\lambda} \\ \underline{0} \end{pmatrix}$ is one solution

Theorem 2.17. *Under Models IIa—VIIa, the least squares estimate of an estimable function* $\underline{\lambda}'\underline{\beta}$ *is given by*

$$\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b},$$

where $\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix}$ *is any solution to the constrained conjugate normal equations.*

Proof. For any solution $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ to the constrained normal equations and any solution $\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix}$ to the constrained conjugate normal equations, we have $X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'\underline{y}$, $A\widehat{\underline{\beta}} = \underline{b}$, $X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda}$, and $A\underline{r}_1 = \underline{0}$.

Now LSE $\underline{\lambda}'\widehat{\underline{\beta}} = (\underline{r}_1'X'X + \underline{r}_2'A)\widehat{\underline{\beta}}$

$= \underline{r}_1'X'X\widehat{\underline{\beta}} + \underline{r}_2'A\widehat{\underline{\beta}}$

$(X'\underline{y} - A'\underline{\rho})$ $\quad \underline{b}$

$= \underline{r}_1'X'\underline{y} - \underline{r}_1'A'\underline{\rho} + \underline{r}_2'\underline{b}$

since $A\underline{r}_1 = 0$

$= \underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b}$

Under Models IIa—VIIa, the least squares estimator of an estimable function $\underline{\lambda}'\underline{\beta}$ is an unbiased estimator. To see this, by Theorem 2.17, the least squares estimator $\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b}$ is of the form $c + \underline{a}'\underline{y}$, with $c = \underline{r}_2'\underline{b}$ and $\underline{a}' = \underline{r}_1'X'$. That is, note

$$c = \underline{r}_2'\underline{b} \qquad \& \qquad \underline{a}'X + \underline{r}_2'A = \underline{r}_1'X'X + \underline{r}_2'A = \underline{\lambda}'$$

match

by Lemma 2.9

because $\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix}$ is solution to constr conjugate NEs

So, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b}$ is unbiased estimator

or $E(\underline{\lambda}'\widehat{\underline{\beta}}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$

Suppose that the two vectors $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable, and let $\widehat{\underline{\beta}}$ represent any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$ (i.e., from a solution to constrained normal equations).

Take $\begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$ and $\begin{bmatrix} S_1 \\ S_2 \end{bmatrix}$ to be any solutions to

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} \Lambda_1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} \Lambda_2 \\ 0 \end{bmatrix}.$$

That is, $X'XR_1 + A'R_2 = \Lambda_1$, $AR_1 = 0$, $X'XS_1 + A'S_2 = \Lambda_2$, and $AS_1 = 0$.

based on solutions to constr. conjugate NEs

Then, by Theorem 2.17, $\Lambda_1'\widehat{\underline{\beta}} = R_1'X'\underline{y} + R_2'\underline{b}$, and $\Lambda_2'\widehat{\underline{\beta}} = S_1'X'\underline{y} + S_2'\underline{b}$.

Under Model IIa, we have $Var(\underline{\epsilon}) = \sigma^2 I = Var(\underline{Y})$

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = R_1'X'(\sigma^2 I)XS_1 = \sigma^2 R_1'X'XS_1 \quad \text{(from above } \Lambda_2 - A'S_2\text{)}$$

$$= \sigma^2 R_1'(\Lambda_2 - A'S_2) = \sigma^2 R_1'\Lambda_2 - \sigma^2(AR_1)'S_2 = \sigma^2 R_1'\Lambda_2,$$

(cov($R_1'X'\underline{Y}, R_2'X'\underline{Y}$))

and

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2(X'XR_1)'S_1 = \sigma^2(\Lambda_1 - A'R_2)'S_1 = \sigma^2\Lambda_1'S_1 - \sigma^2 R_2'(AS_1) = \sigma^2\Lambda_1'S_1.$$

($\Lambda_1 - A'R_2$)

It can also be shown (again, under Model IIa) that, for any generalized inverse $\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$,

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'G_{11}\Lambda_2.$$

ie use $\begin{pmatrix} \hat{\beta} \\ p \end{pmatrix} = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}\begin{pmatrix} X'\underline{Y} \\ \underline{b} \end{pmatrix}$ is a solution to constrained NEs

ie solves $\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\begin{pmatrix} \hat{\beta} \\ p \end{pmatrix} = \begin{pmatrix} X'\underline{Y} \\ \underline{b} \end{pmatrix}$

The proof of the following parallels that of the ordinary Gauss-Markov theorem.

Theorem 2.18. *Under Models IIa—VIIa, if* $\underline{\lambda}'\underline{\beta}$ *is estimable and* $\widehat{\underline{\beta}}$ *is the first component of any solution to the constrained normal equations, then* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is a linear unbiased estimator of* $\underline{\lambda}'\underline{\beta}$*. Under Model IIa, the variance of* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is uniformly less than that of any other linear unbiased estimator of* $\underline{\lambda}'\underline{\beta}$*, that is,* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is the best (minimum variance) linear unbiased estimator (BLUE) of* $\underline{\lambda}'\underline{\beta}$*.*

We can also find an unbiased estimator of σ^2 under Model IIa:

Take $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ to be any solution to the constrained normal equations, and take $\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ to be any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$.

For the residual sum of squares, we have

$$Q(\widehat{\underline{\beta}}) = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}}) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}}) - \widehat{\underline{\beta}}'(X'\underline{y} - X'X\widehat{\underline{\beta}})$$

$$= (\underline{y} - X\widehat{\underline{\beta}})'\underline{y} - \widehat{\underline{\beta}}'A'\underline{\rho} = (\underline{y} - X\widehat{\underline{\beta}})'\underline{y} - \underline{b}'\underline{\rho} = \underline{y}'\underline{y} - \widehat{\underline{\beta}}'X'\underline{y} - \underline{\rho}'\underline{b}$$

$$= \underline{y}'\underline{y} - (G_{11}X'\underline{y} + G_{12}\underline{b})'X'\underline{y} - (G_{21}X'\underline{y} + G_{22}\underline{b})'\underline{b}.$$

It can be shown that, under Model IIa,

$$E[Q(\widehat{\underline{\beta}})] = \sigma^2 \left\{ n - \text{rank} \begin{bmatrix} X \\ A \end{bmatrix} + \text{rank}(A) \right\}.$$

Thus, under Model IIa,

$$\widehat{\sigma}^2 = Q(\widehat{\underline{\beta}})/(n-f) = (\underline{y}'\underline{y} - \widehat{\underline{\beta}}'X'\underline{y} - \underline{\rho}'\underline{b})/(n-f)$$

is an unbiased estimator of σ^2, where $f = \text{rank} \begin{bmatrix} X \\ A \end{bmatrix} - \text{rank}(A)$.

Instead of imposing equality constraints $A\underline{\beta} = \underline{b}$ on the parameter vector $\underline{\beta}$, we might wish to impose q inequality constraints, say

$$A\underline{\beta} \geq \underline{b}.$$

Here, the least squares problem is that of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$ subject to the constraints $A\underline{\beta} \geq \underline{b}$ or, equivalently, that of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$ and $\underline{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_q \end{bmatrix}$ subject to the constraints

$$\begin{aligned} A\underline{\beta} &= \underline{b} + \underline{u}, \\ u_i &\geq 0, \quad i = 1, \ldots, q. \end{aligned}$$

The quantities $u_1, \ldots, u_q$ are called surplus variables.

By taking advantage of our results on the least squares problem with equality constraints, this problem can be reduced essentially to that of minimizing the convex function

$$\left[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\right]' \left[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\right]$$

with respect to $\underline{u}$ subject to the linear equality constraints $(I - AA^-)(\underline{b} + \underline{u}) = \underline{0}$ and the inequality constraints $u_i \geq 0, i = 1, \ldots, q$. This is a quadratic programming problem.

